Kuwait University
College of Life Sciences

Report II: System Requirements Specification

ISC 499 – Capstone Project
Spring 2025/2026

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3-1-2026

Table of Contents

Functional Requirements **4**

Requirement Elicitation techniques and strategies 4

1. Document Analysis 4
2. Supervisor Consultation 4
3. Interviews 5

Survey-Based Validation 6

Functional Modeling 7

1. Use Case Analysis 7

User Stories 13

Data Flow Diagrams 14

Use Case Analysis: Activity diagrams 16

1. Student Query Flow 17
2. GPA Estimation 18
3. Admin Content Management Flow 19

Data Modeling 20

Entity Relationship Diagram: 20

Data Dictionary 21

admin_user 21

Document 21

Source 22

Topic 22

College 22

Chunk 23

User_queries 23

Behavioral Modeling 24

State Transition Diagrams 24

Sequence Diagram 25

Non-Functional Requirements **26**

External Interfaces 26

User Interface 26

Hardware Interfaces 27
Software Interfaces 27
Communications Interfaces 28
Software Requirements 28
1.1. Programming Language and Web Framework 28
2. Large Language Model (LLM) API 28
3. Retrieval-Augmented Generation (RAG) Framework 28
4. Database Management System 29
5. Cloud Hosting Platform 29
6. Version Control System 29
Software Constraints 29
1. Browser Compatibility 29
2. LLM API Dependency 29
3. Arabic Language Rendering 29
4. Internet Connectivity Dependency 30
5. Cloud Platform Dependency 30
Hardware Requirements 30
1. Server-Side Requirements 30
2. Client-Side Requirements 31
Performance Requirements 31
Security Requirements 31
Administrative Access Control 32
Anonymized Query Logging 32
Data Privacy and Session Management 32
Transmission Security and Input Validation 32
Usability Requirements 33
1. Ease of Use 33
2. Bilingual Support 33
3. Accessibility 33
4. Responsive Design 33
Scalability Requirements 34
Appendix A: Survey Questions 35

Appendix B: Survey Results 38

Functional Requirements

Requirement Elicitation techniques and strategies

To ensure that the Chatbot System addresses real academic and administrative needs, multiple requirement elicitation techniques were employed. These techniques were selected to capture both user expectations and institutional constraints while maintaining alignment with the project scope.

1. Document Analysis

Document analysis is a foundational requirement elicitation technique that establishes the baseline for this project. This method was chosen because official university resources provide accurate, verified information and reflect how students currently access services. The following resources were analyzed:

1. The official Kuwait University website and academic portal
2. Official Kuwait University social media channels
3. Existing chatbot systems at other universities (see Related Work in Report I)

Through conducting document analysis, several insights were identified. Information is scattered across multiple platforms, making it difficult for students to find what they need. Critical updates such as schedule changes are only announced through social media accounts rather than official portals. Additionally, the student handbook, a primary resource for students, is only available in Arabic, creating a gap for non-Arabic speaking students. These findings justify the need for a centralized, bilingual, conversational system.

2. Supervisor Consultation

Our ongoing consultations with the project supervisor, Dr. Bader Alkhazi, are being treated as a formal elicitation technique throughout the planning and requirements gathering phases. These sessions are structured around progress reviews, scope clarification, and directional feedback on the system design, and are conducted iteratively in alignment with the project's SDLC model.

Supervisor consultation is included as a formal technique because the supervisor brings both academic expertise and an understanding of the institutional context within which this project

operates. This ensures that the requirements being gathered remain academically realistic within the capstone timeline, and responsive to the standards expected of a formal system specification.

3. Interviews

Structured interviews are to be held with institutional stakeholders whose roles are directly relevant to the chatbot's intended use cases. This technique was chosen because, unlike surveys, interviews allow for open-ended exploration, follow-up questioning, and the possible discovery of implicit requirements. Two stakeholders have been identified for consultation based on their centrality to the academic and administrative experience of Kuwait University students.

Vice Dean of Students' Affairs

The Vice-Dean of Students' Affairs, Dr. Sharifa Alkandari, has been identified as the first consultation given her office's oversight of student-facing administrative services at the institutional level. While students are often directed to Dr. Sharifa's office for matters that require higher-level authorization or administrative escalation in College of Life Sciences, many students also approach her office directly for broader administrative concerns, making her office a comprehensive source of insight into the full range of student administrative needs.

The interview with Dr. Sharifa aims to provide us with insights that will help us to attain categories of student inquiries and cases that her office handles, the policies governing the university information to students, and the expectations around deploying an AI-based student-facing system within Kuwait University's framework. The interview will also explore what level of information the chatbot may appropriately communicate on behalf of the university and where it should redirect students to seek official guidance.

Director of Academic Guidance and Advising Office

The Director of Academic Guidance and Advising Office, Ms. Muneera Al-Marzouq, serves as the frontline point of contact for the majority of student academic inquiries at Kuwait University, with students routinely seeking guidance on course selection, prerequisite clarification, graduation planning, and academic procedures. We aim to have an interview with Ms. Muneera after Dr. Sharifa to cover the same core themes explored with Dr. Sharifa, drawing on her perspective as someone who directly and frequently handles student inquiries on a regular basis. This allows the team to cross-reference insights from both offices, identify any differences in how

student needs are perceived at different levels of the administration, and build a completer and more well-rounded picture of the information gaps the chatbot is designed to address.

Survey-Based Validation

A student survey was conducted to gather broad quantitative input directly from the student population (see Appendix A for survey questions and Appendix B for full survey results). While interviews with institutional stakeholders are still planned to provide deeper qualitative insights, the survey was distributed early to capture student perspectives on information-seeking behavior and chatbot feature preferences across Kuwait University's colleges. Nearly 50 responses were collected from students across multiple colleges, with the majority of respondents from the College of Life Sciences, followed by Engineering and Petroleum, Medicine, Business Administration, and Allied Health Sciences.

The survey results strongly validated the core problem statement of this project. When asked about the challenges they face when searching for university information, the majority of respondents identified information being scattered across multiple platforms as their primary challenge, followed by long processes to find answers, outdated information, and having to contact multiple departments. Furthermore, when asked where they first look for academic or administrative information, students ranked friends and classmates as their most used source, followed by official social media accounts and WhatsApp groups, with the official KU website ranking only fourth. The overall results confirm that students face difficulty finding information and rely more on informal channels for information.

Regarding the proposed chatbot, students expressed strong interest with an average usage likelihood rating of 4.35 out of 5. The most requested topics were registration, campus services, GPA calculation, prerequisite courses, university policies, and equivalent courses, directly validating the core use cases defined in this report. Additionally, students rated source transparency as highly important with an average rating of 4.52 out of 5, confirming the decision to include source references alongside chatbot responses as a key feature.

Functional Modeling

1. Use Case Analysis

The use case analysis presents the functional requirements of the system by modeling interactions between actors and the chatbot. The following components illustrate these interactions and their detailed specifications:

Use Case Diagram:

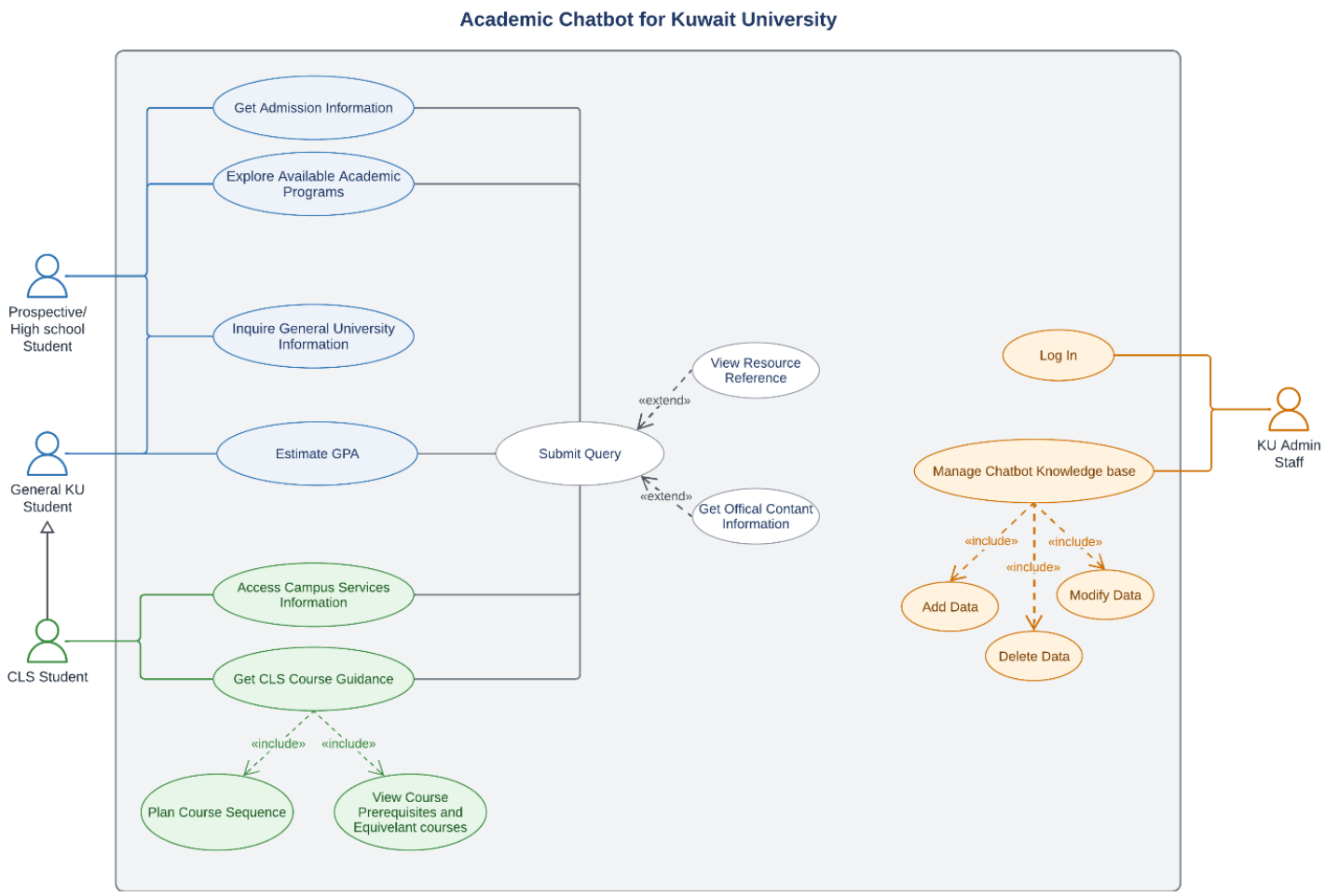


Figure 1: Use Case Diagram

Use Case Specification of Main Use Cases:

Identification	UC01 – Manage Chatbot Knowledge Base
Actors Involved	Primary Actor: Kuwait University Admin Staff
Brief Description	The use case allows authorized staff to manage the knowledge base of the chatbot by adding, modifying, and deleting content to ensure that the content is accurate and up to date.
Assumptions	• The knowledge base stores official academic information. • The system supports authenticated administrative access.
Preconditions	• The admin user is authenticated. • The system is operational and accessible.
Postconditions	• The knowledge base is updated successfully. • The updated information becomes available for chatbot responses.
Priority	High
Frequency of Use	Moderate (annually during academic updates or policy changes).
Main flow of Events	1. Admin logs into the system. 2. The system verifies credentials. 3. Admin accesses the knowledge base management interface. 4. Admin does an operation (Add, Modify, or Delete). 5. Admin enters or edits relevant information. 6. The system validates the input. 7. The system updates the database. 8. The system confirms successful update.
Alternate/Exception Flows	**A1: Invalid Login** 2a. The system displays authentication error message. 2b. The admin retries to login. **A2: Invalid Data Entry** 5a. The system displays validation error 5b. The admin corrects the data entry **E1: System Error** 7a. The system displays an error message. 7b. No changes are saved.
Included Use Cases	• Add Data • Modify Data • Delete Data

Identification	UC02 – Submit Query
Actors Involved	Primary: Prospective Student / KU Student / CLS Student
Brief Description	Allows users to submit queries and receive responses based on official university information.
Assumptions	• Knowledge base contains relevant information.
Preconditions	• System is operational and accessible.
Postconditions	• User receives response. • Interaction logged anonymously.
Priority	High
Frequency of Use	Very High
Main flow of Events	1. User enters query. 2. System processes query. 3. System retrieves relevant information. 4. System generates and sends a response. 5. System logs the query and the response.
Alternate/Exception Flows	**A1: No information found** 3a. Redirect to official contact. **E1: System error** Display error message.
Included Use Cases	• View Resource Reference • Get Official Contact Information

Identification	UC03 – Get Admission Information
Actors Involved	Primary: Prospective / High School Student
Brief Description	The use case provides prospective students with admission requirements and procedures.
Assumptions	• Admission data exists in the knowledge base.
Preconditions	• The system is operational and accessible. • The user can access the chatbot

Postconditions	• The student receives relevant admission details.
Priority	High
Frequency of Use	High (during admission periods)
Main flow of Events	1. Student submits admission query. 2. System retrieves official data. 3. System displays response.
Alternate/Exception Flows	**A1: No information found** 2a. Redirect to official contact. **E1: System error** Display error message.
Included Use Cases	Submit Query

Identification	**UC04 – Explore Available Academic Programs**
Actors Involved	Primary Actor: Prospective Student
Brief Description	This use case enables students to explore available academic programs and related information about the program and their academic future.
Assumptions	• Program information is available in the knowledge base.
Preconditions	The system is operational.
Postconditions	The student receives program descriptions and requirements.
Priority	Medium
Frequency of Use	Moderate
Main flow of Events	1. The student sends a query about available programs. 2. The system retrieves program information. 3. The system displays a response with appropriate program details.
Alternate/Exception Flows	**A1: No information found** 2a. Redirect to official contact. **E1: System error** Display error message.
Included Use Cases	Submit Query

Identification	UC05 – Estimate GPA
Actors Involved	Primary: General Kuwait University Student
Brief Description	This use case allows students to estimate their GPA based on manually entered course grades and credits.
Assumptions	GPA calculation formula is predefined in the system.
Preconditions	• The system is operational. • The student provides course grades and credit hours.
Postconditions	The estimated GPA is displayed to the student. No personal academic data is permanently stored.
Priority	Medium
Frequency of Use	Moderate (During exam periods)
Main flow of Events	1. Student selects GPA estimation. 2. Student enters grades. 3. System calculates GPA. 4. System displays result and gives suggestion.
Alternate/Exception Flows	**A1: No information found** 2a. Redirect to official contact. **E1: System error** Display error message.
Included Use Cases	Submit Query

Identification	UC06 – Access Campus Services Information
Actors Involved	Primary: College of Life Science Student
Brief Description	Provides students with information about campus services that are available in College of Life Science.
Assumptions	Service information of CLS is available on the knowledge base.
Preconditions	The system is operational.
Postconditions	The student receives specific service details at College of Life Science.
Priority	High

Frequency of Use	High (during the beginning of the semester)
Main flow of Events	1. Student submits service query related to CLS. 2. System retrieves relevant service information. 3. System displays response.
Alternate/Exception Flows	**A1: No information found** 2a. Redirect to official contact. **E1: System error** Display error message.
Included Use Cases	Submit Query

Identification	**UC07 – Get CLS Course Guidance**
Actors Involved	Primary: College of Life Science Student
Brief Description	This use case enables CLS students to receive course planning guidance and prerequisite information.
Assumptions	CLS program data exists in the knowledge base.
Preconditions	The system is operational.
Postconditions	The student receives course guidance response.
Priority	High (within College of Life Science scope)
Frequency of Use	High (during registration period)
Main flow of Events	1. Student submits CLS query. 2. System retrieves program structure. 3. System displays prerequisites and sequence or course specific guidance.
Alternate/Exception Flows	**A1: Missing course data** Redirect to advising office or academic advisor.
Included Use Cases	1. Submit Query 2. Plan Course Sequence 3. View Course Prerequisites

User Stories

User stories capture the functional expectations of each actor identified in the Use Case Diagram, ensuring that the system's features are driven by real user needs across all target groups from primary students like prospective students exploring Kuwait University for the first time, to enrolled KU students for general KU information and CLS students managing their academic plans, to secondary users, i.e., administrative staff maintaining the chatbot's knowledge base.

- **High School Students / Prospective**

  1. As a prospective student, I want to access admission requirements and application details so that I can prepare properly for university admission.
  2. As a prospective student, I want to explore available academic programs and their descriptions so that I can choose a suitable major.

- **General Kuwait University Student**

  1. As a Kuwait University student, I want to ask general university-related questions so that I can quickly obtain accurate academic information.
  2. As a Kuwait University student, I want to estimate my GPA so that I can plan my academic performance and future study decisions.

- **College of Life Science (CLS) Student**

  1. As a CLS student, I want to receive course guidance specific to my major so that I can make informed registration decisions.
  2. As a CLS student, I want to plan my course sequence so that I can complete program requirements efficiently.
  3. As a CLS student, I want to view course prerequisites and equivalent courses so that I can avoid registration errors.
  4. As a CLS student, I want to access campus services information so that I can make use of available university facilities and resources.

- **All Primary Users**

1. As a user, I want responses to be based on official university sources so that I can trust the information provided.
2. As a user, I want to be redirected to the appropriate official department when the chatbot cannot answer my question so that I can obtain accurate assistance.
3. As a user, I want to view the source reference behind the chatbot's response so that I can verify the information.

- **Secondary Users: Kuwait University Administration Staff**

1. As a KU administrator, I want to add, modify, and delete content in the chatbot's knowledge base so that the information provided to students remains accurate and up to date.

Data Flow Diagrams

In this section, two Data Flow Diagrams (DFDs) are presented using Yourdon and Coad's notation to show how data moves through the Academic Chatbot system. The Level 0 diagram provides a high-level overview of the system and its interactions with external entities, while the Level 1 diagram zooms into the internal processes of the system and details how data flows between them.

Level 0 – Context Diagram

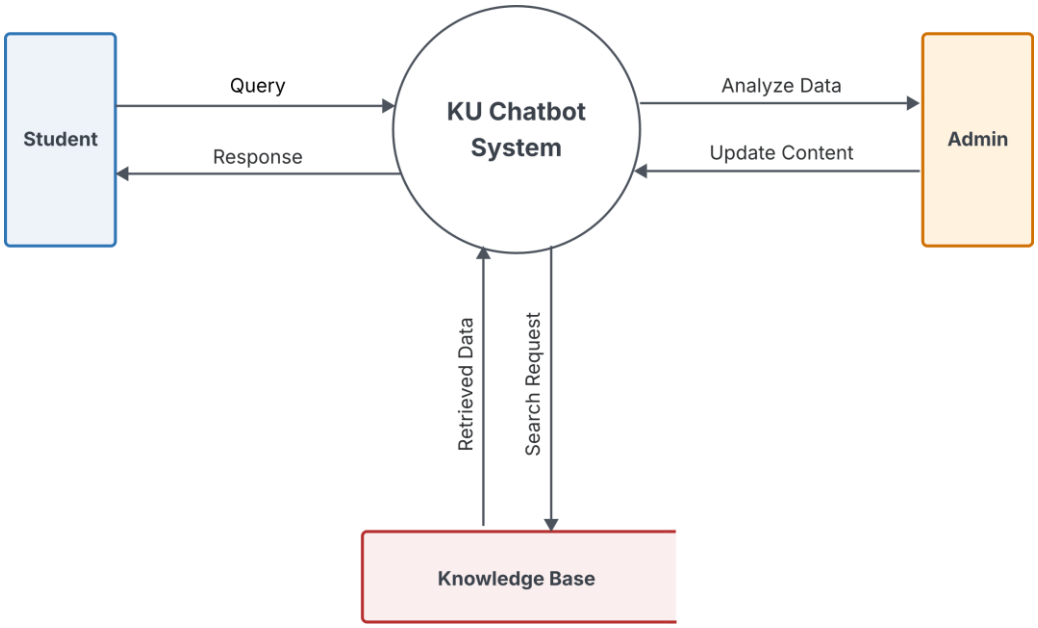


Figure 2: Level 0 Context Data Flow Diagram

Level 0 (L0) shows two main external entities that interact with the system:

- **Students** (Prospective, General KU, and CLS students): Students submit queries to the chatbot and receive responses in return.
- **KU Administrative Staff**: Staff submit content updates to maintain the knowledge base and receive query logs and analyzed data to support system improvements.

The system interacts with a **Knowledge Base** data store, retrieving relevant information to answer queries and storing updated content following administrative actions.

Level 1 – Detailed Data Flow

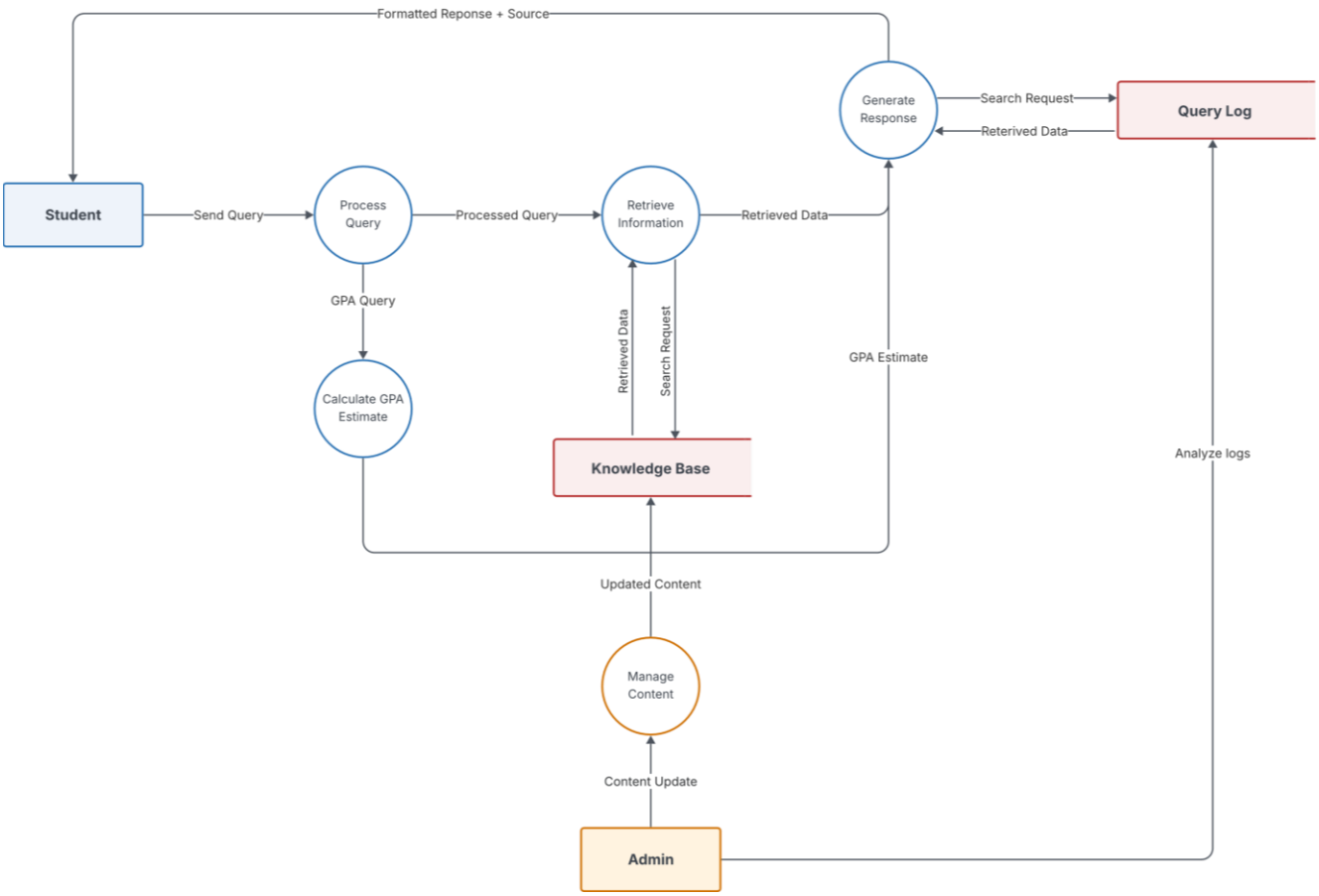


Figure 3: Level 1 Detailed Data Flow Diagram

The Level 1 (L1) Data Flow Diagram expands on the Context Diagram (L0) by breaking the system

into five internal processes, revealing how data moves through the chatbot internally. These processes are mentioned below:

- **Process Query:** The entry point of the system. When a student submits a query, it is received and routed accordingly. General academic queries are directed to the Retrieve Information process, while GPA-related queries are routed to the Calculate GPA Estimate process.
- **Retrieve Information:** Upon receiving a query, this process searches the Knowledge Base and retrieves the relevant data along with its source reference, which is then passed to the Generate Response process.
- **Generate Response:** This process takes the retrieved information and formats it into a response that is returned to the student. It also logs the query and response anonymously into the Query Log data store.
- **Calculate GPA Estimate:** This process handles GPA queries independently without interacting with the Knowledge Base. It receives grade input from the student and returns a calculated GPA estimate directly.
- **Manage Content**: This process is triggered exclusively by administrative staff. It receives content updates from the admin and applies them to the Knowledge Base. The admin also receives query logs and analyzed data from the Query Log to support knowledge base improvements.

Use Case Analysis: Activity diagrams

Activity diagrams are used to model the dynamic behavior of the system by illustrating the flow of activities and actions. The following diagrams represent the key workflows within the Chatbot system:

1. Student Query Flow

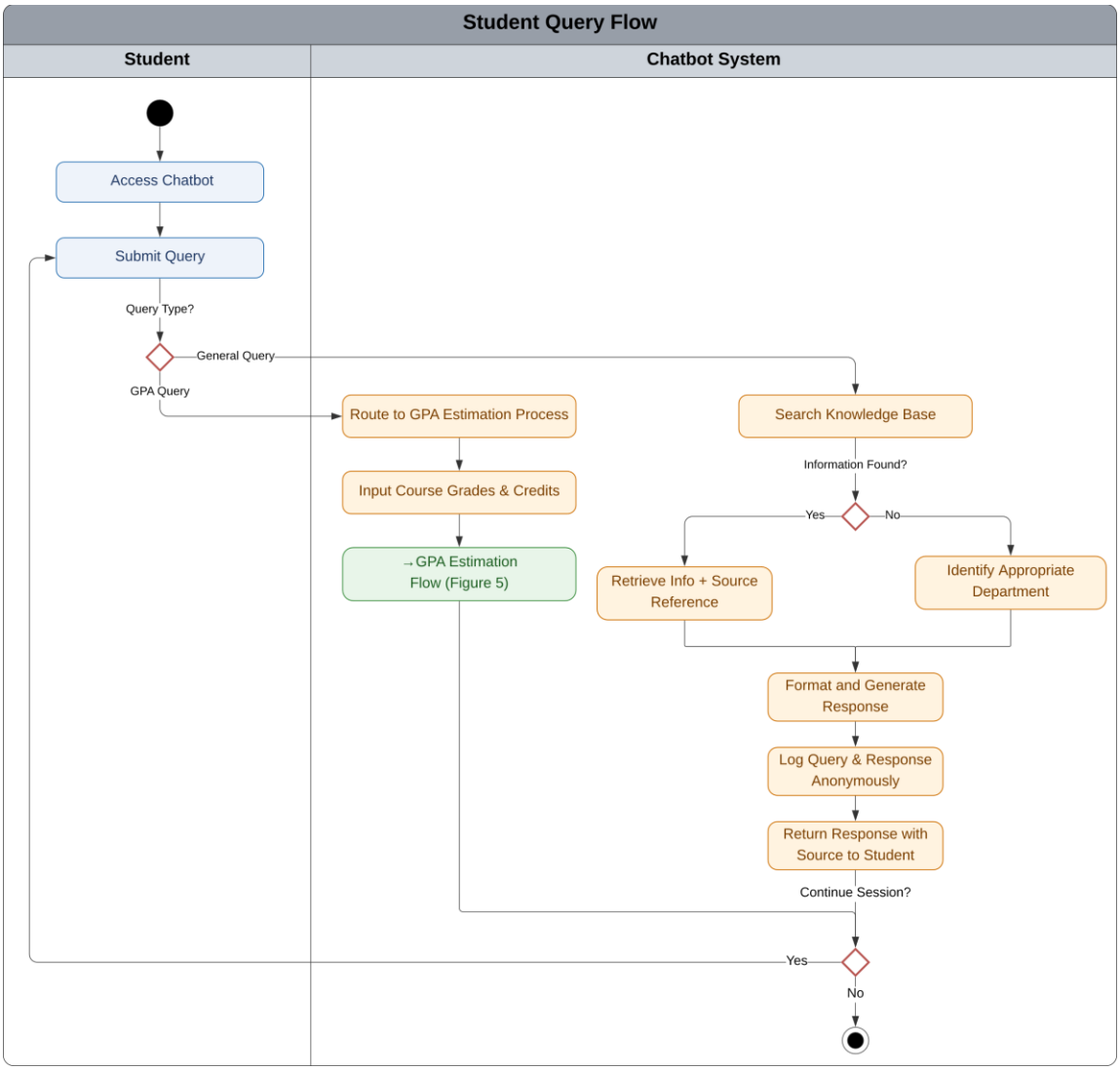


Figure 4: Student Query Flow Activity Diagram

The diagram above illustrates the main interaction between students and the chatbot. The flow begins when a student enters and submits a query. This query can be classified into two types: GPA estimation, which does not require referring to the knowledge base and is handled through a separate process shown in Figure 5, and general queries, which require searching the knowledge base for relevant information where the system retrieves matching information and generates a response with source citation. If no match is found, a fallback message is displayed with the appropriate department to refer to. The student can then choose to continue asking questions or end the session.

2. GPA Estimation

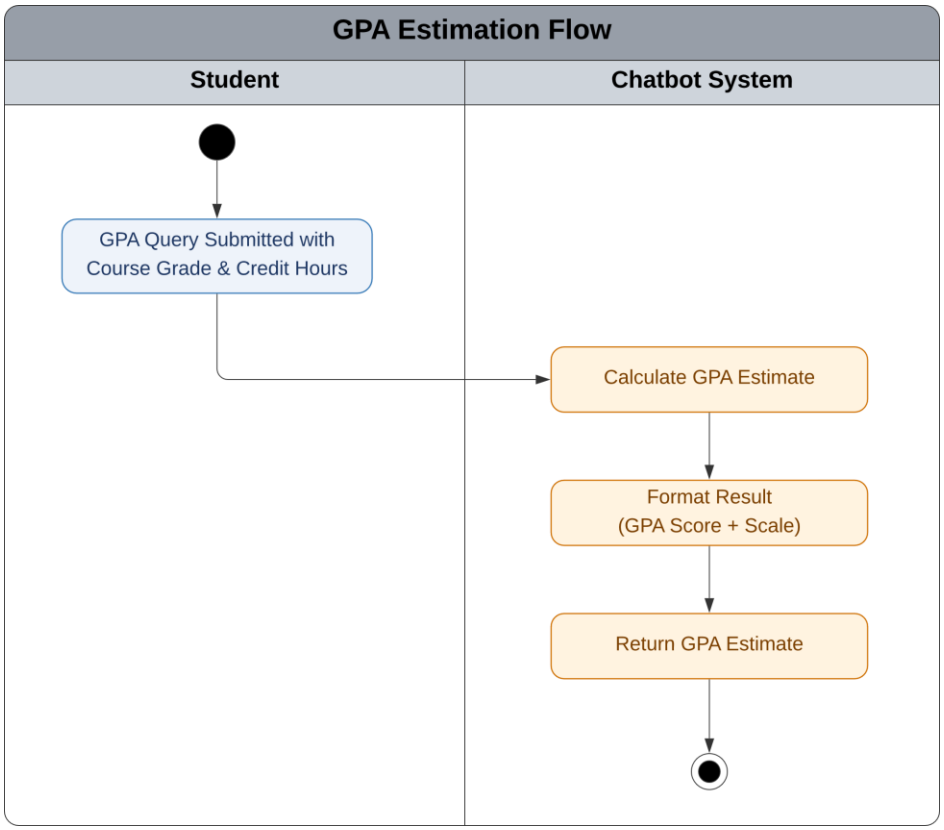


Figure 5: GPA Estimation Flow Activity Diagram

This diagram illustrates the GPA estimation feature. This flow is triggered when a student's query is identified as a GPA-related request. The student is prompted to enter their course grades and credit hours, and the system calculates, formats, and displays the estimated GPA.

3. Admin Content Management Flow

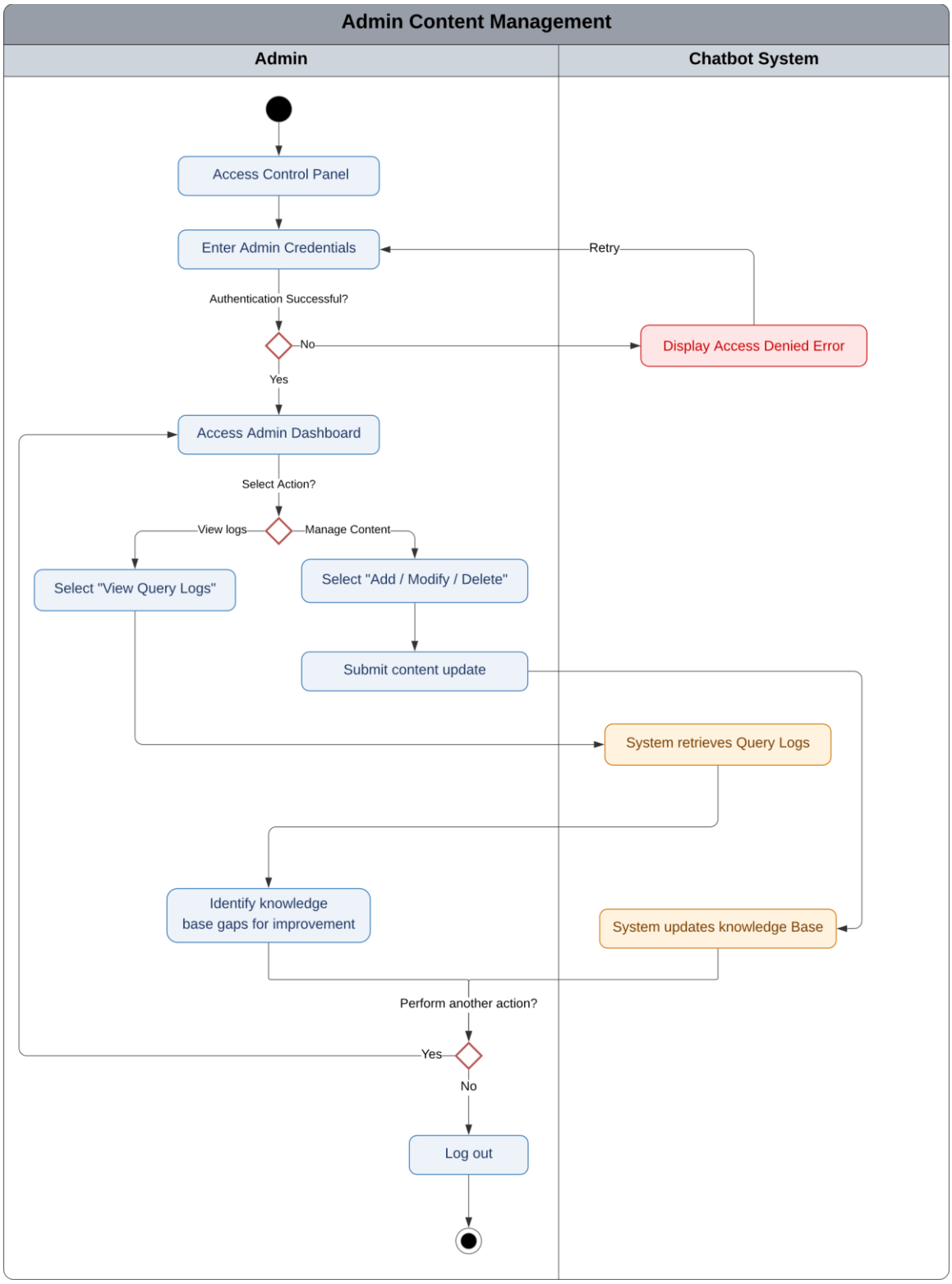


Figure 6: Admin Content Management Flow Diagram

The above diagram shows the workflow of an authorized Kuwait University administrator interacting with the chatbot's control panel. The process begins with the admin navigating to the panel and entering their credentials, with failed authentication prompting a retry. Upon successful login, the admin can either manage knowledge base content by adding, modifying, or deleting entries or view anonymized query logs to identify gaps and inform future content improvements. Finally, the admin can then perform another action or log out to end the session.

Data Modeling

Entity Relationship Diagram:

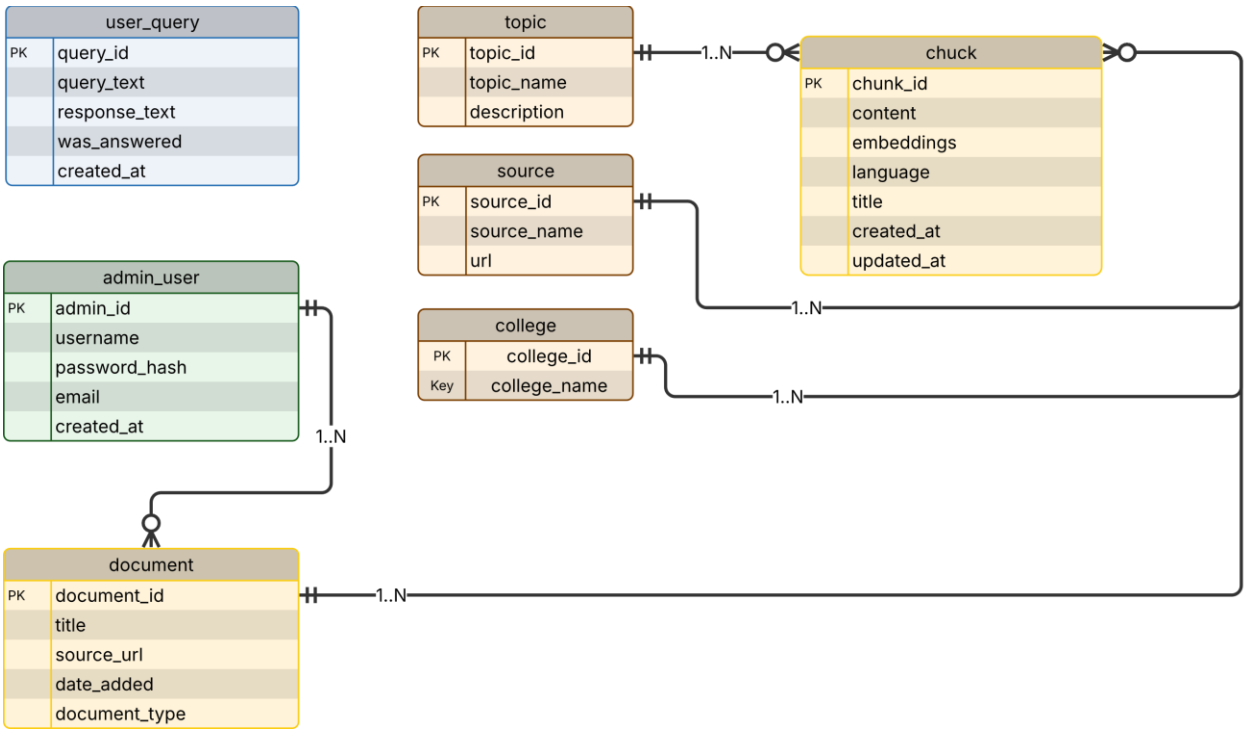


Figure 7: Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the conceptual data model for the KU AI Chatbot system. The diagram consists of the following seven entities, described below, that work together to support the chatbot's core functionalities:

- **admin_user:** stores the credentials for administrators who manage the chatbot system, including username, hashed password, email, and account creation timestamp.

- **document:** tracks the original files uploaded to the system, such as PDFs, Word documents, or web pages, with each document having a title, source URL, upload date, and document type. This provides traceability for all content added to the knowledge base.
- **source:** maintains a list of official sources from which information is collected, such as the KU official website, student handbook, or academic guides.
- **topic:** categorizes content by topic, such as Registration, Admission, or Policies.
- **college:** stores the list of colleges at Kuwait University, enabling the system to filter and retrieve college-specific information.
- **chunk:** is the central entity of the system. When documents are processed, they are split into smaller text pieces called chunks. Each chunk contains the text content, a title, language indicator, and an embedding field that stores the vector representation used by the RAG system to find semantically similar content when answering user queries.
- **user_query:** logs all user interactions with the chatbot, storing the original query text, the response generated, whether the query was successfully answered, and a timestamp. This entity stands alone as it is used for analytics purposes only.

Data Dictionary

admin_user

Field	Data Type	Constraints	Description
admin_id	INT	PK	Unique identifier for each admin
username	TEXT	NOT NULL, UNIQUE	Admin login username
password_hash	TEXT	NOT NULL	Encrypted password for security
email	TEXT	NOT NULL, UNIQUE	Admin email for account recovery
created_at	TIMESTAMP		Account creation date and time

Relationship:

- One admin can add and edit many documents (N:1)

Document

Field	Data Type	Constraints	Description
document_id	INT	PK	Unique identifier for each document
title	TEXT	NOT NULL	Name of the uploaded document
source_url	TEXT		Original URL of the document if applicable

date_added	TIMESTAMP	NOT NULL	Date and time the document was uploaded
document_type	TEXT		Type of document (PDF, DOCX, HTML, etc.)

Relationship:

- Many documents can be added by one admin (N:1)
- One document can belong to many chunks (1:N)

Source

Field	Data Type	Constraints	Description
source_id	INT	PK	Unique identifier for each source
source_name	TEXT	NOT NULL	Name of the source (e.g., KU Website, Student Handbook)
url	TEXT		URL link to the source

Relationship:

- One source can belong to many chunks (1:N)

Topic

Field	Data Type	Constraints	Description
topic_id	INT	PK	Unique identifier for each topic
topic_name	TEXT	NOT NULL	Name of the topic (e.g., Registration, Admission)

Relationship:

- One topic can belong to many chunks (1:N)

College

Field	Data Type	Constraints	Description
college_id	INT	PK	Unique identifier for each college
college_name	TEXT	NOT NULL	Name of the college (e.g., College of Life Sciences)

Relationship:

- One college can belong to many chunks (1:N)

Chunk

Field	Data Type	Constraints	Description
chunk_id	INT	PK	Unique identifier for each chunk
content	TEXT	NOT NULL	The text content of the chunk
title	TEXT		Title or heading of the chunk

language	TEXT		Language of the content (Arabic or English)
embedding	VECTOR		Vector representation for semantic search
created_at	TIMESTAMP		Date and time the chunk was created
updated_at	TIMESTAMP		Date and time the chunk was last updated

Relationships:

- Many chunks belong to one document (N:1)
- Many chunks belong to one source (N:1)
- Many chunks belong to one topic (N:1)
- Many chunks belong to one college (N:1)

User_queries

Field	Data Type	Constraints	Description
query_id	INT	PK	Unique identifier for each query
query_text	TEXT	NOT NULL	The user's query
response_text	TEXT		The chatbot's generated response
was_answered	BOOLEAN	NOT NULL	Whether the query was successfully answered or not
created_at	TIMESTAMP		Date and time the query was made

Behavioral Modeling

Behavioral modeling describes how the system responds to events and transitions between different states over time. Unlike functional modeling, which focuses on what the system does, behavioral modeling captures how the system behaves dynamically during interactions. In this section, two types of behavioral diagrams are presented: **State Transition Diagrams**, which show the states that the system moves through during a session, and a **Sequence Diagram**, which will show the order of interactions between system components.

State Transition Diagrams

State Transition Diagram – User Query

This diagram models the behavior of the chatbot system during a student interaction session. To summarize the behavioral flow, key steps are mentioned below:

1. The system begins in an Idle state and transitions to Waiting for Input once the user opens the chatbot.
2. Upon query submission, the system enters the Processing Query state and routes the query based on its type: general queries proceed to Retrieving Knowledge, while GPA-related queries are directed to Calculating GPA.
3. Once knowledge is retrieved or the GPA is calculated, the system transitions to Generating Response and subsequently to Displaying Response.
4. From the Displaying Response state, the user may either submit another query, returning the system to Waiting for Input, or exit the session, transitioning the system to Session Ended.

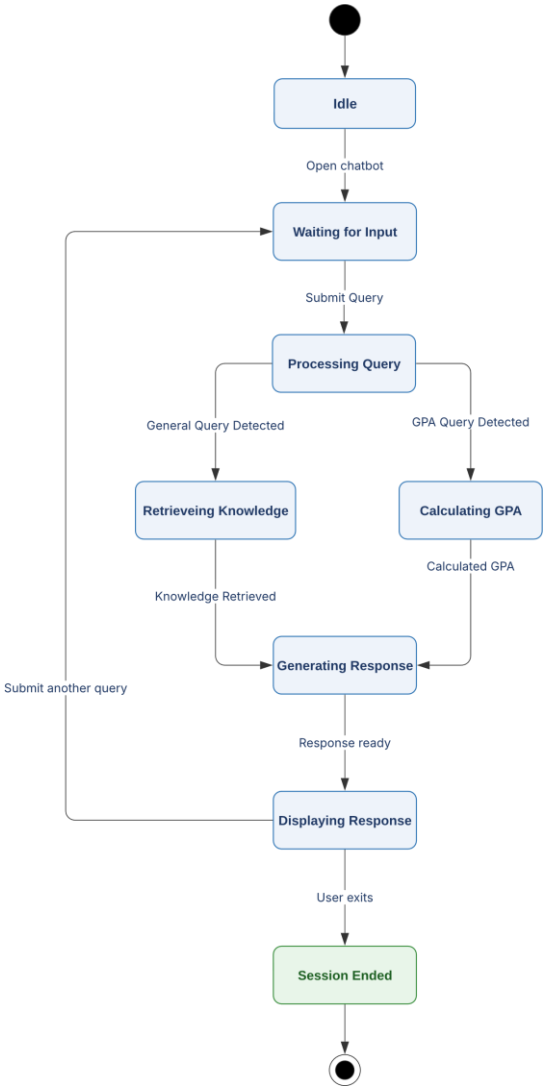


Figure 8: State Transition Diagram – User Query

State Transition Diagram – Admin Control Panel

The diagram models the behavior of the administrative control panel. This behavioral modeling is described as follows:

1. The system begins in an Idle state and transitions to Authenticating once the admin navigates to the panel and enters credentials.
2. Invalid credentials transition the system to Authentication Failed, from which the admin can retry.
3. Upon successful authentication, the system transitions to Control Panel, from which the admin can select content management which transitions the system through Managing Content, Updating Knowledge Base, and Update Confirmed before returning to Control Panel, or the admin may select query log which transitions the system to Viewing Query Logs and subsequently to Reviewing Logs.
4. The admin can perform further actions or log out, transitioning the system to Session Ended.

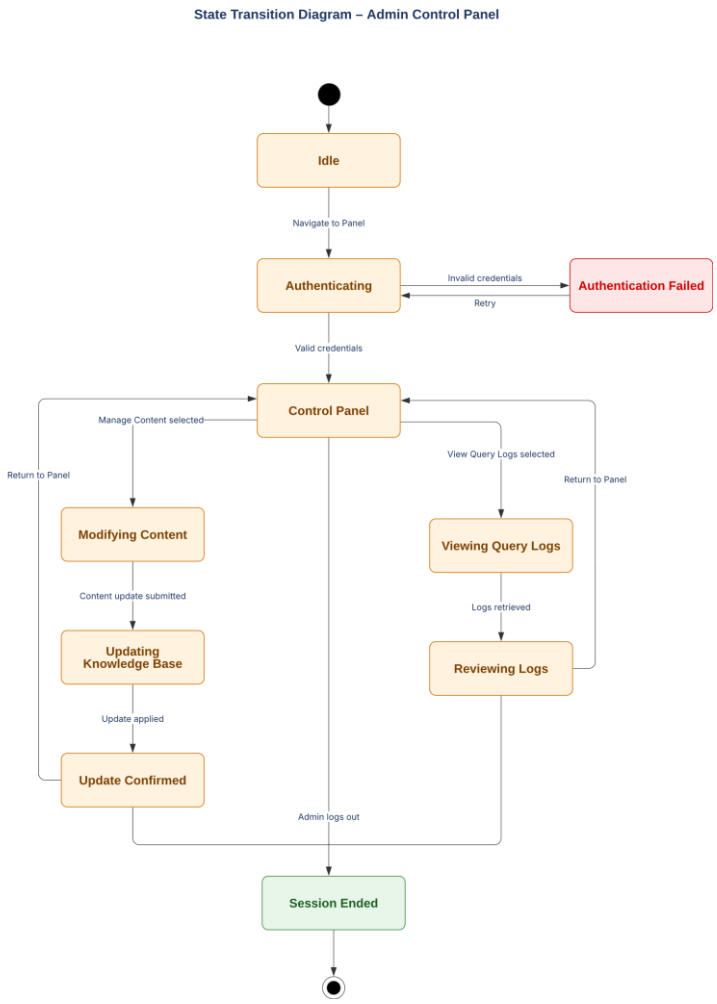


Figure 9: State Transition Diagram – Admin Control Panel

Sequence Diagram

The sequence diagram illustrates the interaction flow between the Student, Chatbot, and Knowledge Base when processing an academic query. The flow begins when a student submits a query, which the chatbot processes by searching the knowledge base for relevant information. Based on the results, the chatbot either returns the answer with a source citation or identifies the appropriate department to refer the student to.

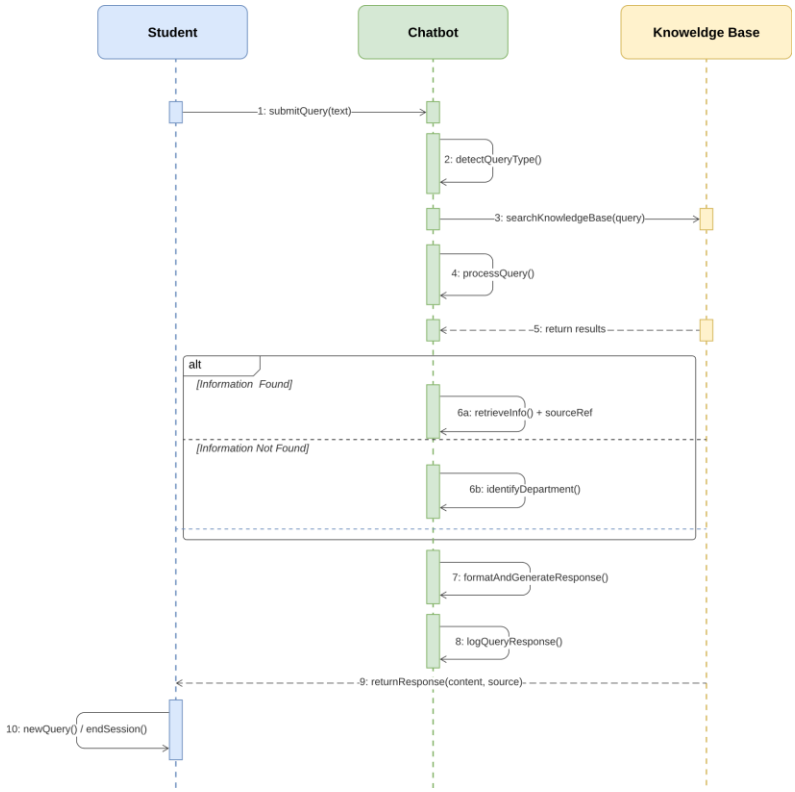


Figure 10: Sequence Diagram for Student Query

Non-Functional Requirements

External Interfaces

User Interface

The system provides web-based interfaces corresponding to its primary and secondary users. The student-facing interface is a conversational chat application accessible through any modern browser on desktop or mobile devices. The interface follows a standard chat layout with a message input field at the bottom and a scrollable conversation history above. All screens will display a consistent header identifying the chatbot as an official Kuwait University tool. The interface supports full bilingual display in Arabic and English, with automatic right-to-left rendering for Arabic text. Source references are displayed with responses. Error messages, such as those triggered by unrecognized queries or connectivity issues, will be displayed inline within the chat in a clearly distinguishable format.

The administrative interface is a separate secured panel accessible only to authorized Kuwait University staff following credential-based authentication. It provides clearly labeled controls for adding, modifying, and deleting knowledge base content, as well as a dedicated section for reviewing anonymized query logs. Error messages for failed authentication will be displayed prominently within the panel.

Detailed UI design specifications will be documented in a separate user interface specification during the system design phase.

Hardware Interfaces

The system does not require direct interaction with specialized hardware components. On the client side, standard input devices including keyboards, touchscreens, and microphones for voice input are supported through the web browser. The system is designed to function across a range of devices including desktops, laptops, smartphones, and tablets. On the server side, all hardware interaction is abstracted through the selected cloud hosting platform, with no direct physical hardware management required.

Software Interfaces

The system interfaces with the following external software components:

- **Large Language Model (LLM) API:** The system communicates with an LLM API to process natural language queries and generate context-aware responses. Query text is sent to the API as input, and generated response text is received as output.
- **Knowledge Base:** The system retrieves relevant information chunks from a structured knowledge base containing static university data and dynamically updated content from official sources. Data flows from the knowledge base into the retrieval pipeline, where it is injected into the LLM prompt as context before response generation.
- **Cloud Hosting Platform:** The system is currently planned to be deployed on a cloud platform that provides the operating environment for the web application, backend server, and database.

Communications Interfaces

All communication between the client and the server is conducted over HTTPS, ensuring encrypted and secure data transmission in compliance with standard web security practices. A stable internet connection is required on both the server and client sides to support real-time query processing and API communication. Specific data transfer rates and synchronization mechanisms will be defined during the system design phase in accordance with the selected cloud platform and API provider requirements.

Software Requirements

The following software requirements define the categories of tools, technologies, and services necessary for the development and deployment of the Academic Chatbot system. Specific technology selections will be finalized during the system design phase in accordance with project requirements and feasibility.

1. **1. Programming Language and Web Framework**

The system requires a suitable programming language and web development framework to support both the frontend user interface and the backend server logic. The selected language must support Arabic and English text processing, and communication with external APIs. The framework must be capable of rendering a responsive, bilingual web interface compatible across modern browsers and devices.

2. **Large Language Model (LLM) API**

The chatbot's response generation relies on a third-party Large Language Model API. The selected API must support natural language understanding, context-aware response generation, and bilingual processing in both Arabic and English.

3. **Retrieval-Augmented Generation (RAG) Framework**

The system aims to implement a RAG architecture to ground the chatbot's responses in verified knowledge base content. A RAG framework or supporting library will be required to manage the retrieval pipeline, including query embedding, similarity search, and context injection into the LLM prompt.

4. Database Management System

A database management system is required to store and manage the knowledge base content, anonymized query logs, and administrative user credentials.

5. Cloud Hosting Platform

A cloud hosting platform is required to deploy and serve the web application, backend server, and database, providing reliable uptime, scalable resource allocation, and HTTPS support.

6. Version Control System

A version control system is required to manage the codebase throughout the development lifecycle, supporting collaborative development, change tracking, and code backup.

Software Constraints

The following constraints define the technical software-specific limitations within which the system must operate.

1. Browser Compatibility

The chatbot is designed to function on all modern web browsers including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. The system does not guarantee full compatibility with outdated or unsupported browser versions, and users are advised to use an up-to-date browser for the best experience.

2. LLM API Dependency

The chatbot's core response generation relies on a Large Language Model API. Any downtime, rate limiting, or changes to the API's terms of service by the provider directly impacts the system's availability and response quality. The system's performance is therefore partially dependent on the reliability and continued availability of the selected API provider.

3. Arabic Language Rendering

Full Arabic language support requires proper Unicode encoding and right-to-left text rendering in the chosen web development framework. The interface must correctly handle bidirectional text display, ensuring that Arabic and English content are rendered appropriately within the same interface without formatting conflicts.

4. Internet Connectivity Dependency

The system requires a stable internet connection on both the server and client sides. On the server side, connectivity is required to communicate with the LLM API and retrieve dynamic knowledge base updates. On the client side, users must have an active internet connection to access and interact with the chatbot.

5. Cloud Platform Dependency

The system's deployment is dependent on the selected cloud hosting platform. Any service interruptions, configuration changes, or policy updates from the cloud provider may affect system availability.

Hardware Requirements

This chatbot system is a web-based application, and so the hardware requirements are considered from two perspectives: the server-side infrastructure responsible for hosting and running the system, and the client-side devices through which users will access it.

1. Server-Side Requirements

Cloud hosting has been selected for this project as it provides on-demand resource provisioning, scalability, and reliability without requiring dedicated physical infrastructure. The chatbot will require a cloud environment capable of supporting the following concurrently:

- A web application interface
- A backend server
- A database
- An AI-based retrieval system

Various platforms are being considered, with the final selection to be confirmed during the system design phase. The cloud environment will be configured to meet the processing, storage, and bandwidth demands of an AI-based retrieval system handling concurrent user sessions, with specific resource allocation to be defined at that stage.

Should the system expand beyond its initial scope to serve a larger user base or additional colleges across Kuwait University, the cloud infrastructure can be scaled up to meet increased demand without requiring a complete system overhaul.

2. Client-Side Requirements

No specialized hardware is required from the user. The system is designed to be accessible through any modern device with a web browser and an internet connection, including desktop computers, laptops, smartphones, and tablets. This ensures broad accessibility across the student population without placing any hardware burden on the end user, which is particularly important given the system's goal of inclusive and equitable access.

Performance Requirements

Performance requirements define the expected operational efficiency of the chatbot system. These requirements ensure that the system performs reliably under normal and peak academic usage conditions:

- **Response Time:** Rapid generation of responses to ensure a prompt and efficient user experience. Simple queries shall be processed within a few seconds, while complex queries requiring deeper knowledge base retrieval may take additional time.
- **Accuracy:** High precision in understanding and addressing student queries, with source citations provided for verified information.
- **Throughput:** Ability to handle multiple concurrent users during prototype testing, with stable session management throughout active conversations.
- **Adaptability:** The knowledge base structure allows for continuous updates and expansion to accommodate evolving student needs and new information.
- **Reliability:** Dependable performance with 24/7 availability, ensuring the chatbot is accessible when needed.

Security Requirements

The chatbot system is designed as an open-access platform for current and prospective students, meaning no login or authentication is required for general use. However, security considerations remain critical given that the system handles academic information, processes user queries, and includes an authenticated administrative control panel with content management capabilities. The following security requirements have been defined to ensure the system operates safely and responsibly.

Administrative Access Control

The administrative control panel requires secure authenticated access restricted to authorized Kuwait University staff. This ensures:

- Only approved personnel can add, modify, or delete knowledge base content and review query logs
- Administrative privileges are explicitly granted and inaccessible to general users
- Admin credentials will be managed and maintained by authorized university staff

Anonymized Query Logging

To support continuous system improvement, anonymized query logs will be maintained. Each log entry records only the query submitted, the response generated, and a timestamp, with no personally identifiable information captured or linked to any entry. These logs will be used solely to identify gaps in the knowledge base to improve responses. Access to query logs will be restricted exclusively to authorized administrative personnel through the admin control panel.

Data Privacy and Session Management

Since the chatbot operates without student authentication, no personal student data is collected, stored, or retained. This ensures:

- No personal student information is stored or exposed
- GPA inputs and query content are never tied to any individual
- Users are advised not to include personally identifiable information such as their name or student ID in their queries, as the system is not responsible for personal information voluntarily included in submitted queries

Transmission Security and Input Validation

All data transmitted between the user's device and the system will be secured through HTTPS encryption, protecting query content and system responses from interception. Backend systems and the knowledge base will be protected through standard cloud security configurations provided by the selected hosting platform.

Usability Requirements

The chatbot system is designed to be intuitive, accessible, and inclusive for all users across Kuwait University's diverse student population. The following usability requirements have been defined to ensure a consistent and equitable user experience.

1. Ease of Use

The interface is designed to require minimal prior experience or technical knowledge to operate. Students should be able to submit a query and receive a response without navigating through multiple steps or menus. The conversational interface follows familiar chat-based interaction patterns, reducing the learning curve for first-time users.

2. Bilingual Support

The system intends to provide full and equal support for both Arabic and English. The interface accommodates both languages simultaneously, and the chatbot aims to automatically detect the language of the user's input and respond accordingly.

3. Accessibility

The system is designed to be accessible to users with diverse accessibility needs through the following features:

- **Voice Input**: Users can interact with the chatbot using voice input as an alternative to typing, ensuring accessibility for users who may have difficulty with text-based interaction.
- **Screen Reader Compatibility**: The interface is designed to be compatible with standard screen readers, ensuring that visually impaired users can navigate and interact with the system effectively.

4. Responsive Design

The interface is fully responsive and adapts to different screen sizes and device types, including desktops, laptops, smartphones, and tablets, ensuring a consistent experience regardless of the device used.

Scalability Requirements

This section outlines the scalability requirements to accommodate future growth in user load and data volume in the chatbot:

- The system shall support a minimum number of concurrent users during the prototype phase without performance degradation, with the ability to scale and accommodate all Kuwait University students in the future.
- The architecture must allow horizontal scaling to support university-wide deployment.
- The knowledge base will be structured by college and topic, enabling expansion to include more colleges and extra information without affecting the performance.
- The system will follow a modular architecture, allowing new features and integrations to be added with minimal impact on existing functionality.

Appendix A: Survey Questions

Kuwait University Chatbot Survey 🤖

This survey explores the need for a university chatbot that makes finding information easier. Your responses are anonymous and will be used for academic purposes only.
يهدف هذا الاستبيان إلى تقييم الحاجة لروبوت محادثة (Chatbot) جامعي يسهّل الوصول إلى المعلومات. جميع الإجابات ستستخدم لأغراض أكاديمية فقط.
للإجابة على الاستبيان باللغة العربية، يرجى الضغط على رمز 🌐 لتغيير اللغة.

* Required

Section 1: General Questions

1. Which College are you in? *

- ◯ Law
- ◯ Arts
- ◯ Science
- ◯ Life Sciences
- ◯ Medicine
- ◯ Engineering & Petroleum
- ◯ Business Administration
- ◯ Allied Health Sciences
- ◯ Education
- ◯ Sharia and Islamic Studies
- ◯ Social Sciences
- ◯ Pharmacy
- ◯ Dentistry
- ◯ Architecture
- ◯ Medical Sciences Center
- ◯ Faculty of Public Health

2. Which year are you in? *

- ◯ First year
- ◯ Second year
- ◯ Third year
- ◯ Fourth year
- ◯ Fifth year or above

3. When you need academic or administrative information, where do you usually look first?

- ◯ KU official website
- ◯ Instagram accounts
- ◯ WhatsApp groups
- ◯ Friends / classmates
- ◯ Academic advisor
- ◯ Other

4. How often do you struggle to find accurate or updated university information?

1	2	3	4	5

5. What challenges do you face when searching for university information?

- ◯ Information is scattered across multiple platforms
- ◯ Outdated information
- ◯ Refer to multiple departments
- ◯ Takes too long to find answers
- ◯ Difficult to understand formal language
- ◯ Only available in Arabic
- ◯ Other

6. Do you find it easier to keep up with university related information from official social media accounts rather than official website?

- ◯ Yes
- ◯ No
- ◯ Maybe

7. If there was a chatbot that would help answer your questions regarding ... how often would you use it

1	2	3	4	5

8. Which topics would you most want the chatbot to answer?

- ◯ Registration
- ◯ Admission
- ◯ Prerequisites
- ◯ Campus services
- ◯ Policies
- ◯ GPA calculation
- ◯ Other

9. if the chatbt can estimate your gpa by just sending hm your grades Would you find this feature useful?

- ◯ No
- ◯ Yes
- ◯ Maybe

10. on a scale of 1-5 When you use chatbots do you prefer having refernce to the source it brought the information from if needed?

1	2	3	4	5

Appendix B: Survey Results

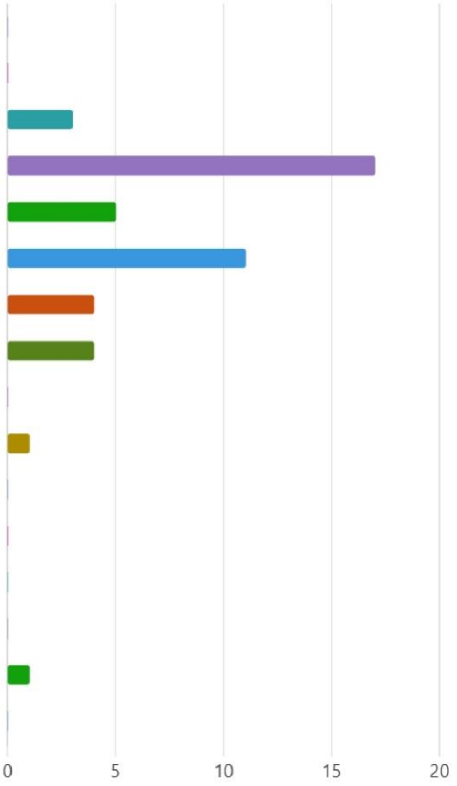
Responses Overview

Active

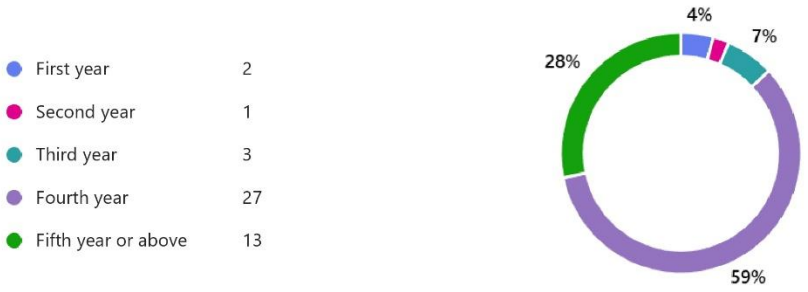
Responses	Average Time	Duration
45	06:19	1 Days

1. Which College are you in?

Law	0
Arts	0
Science	3
Life Sciences	17
Medicine	5
Engineering & Petroleum	11
Business Administration	4
Allied Health Sciences	4
Education	0
Sharia and Islamic Studies	1
Social Sciences	0
Pharmacy	0
Dentistry	0
Architecture	0
Medical Sciences Center	1
Faculty of Public Health	0

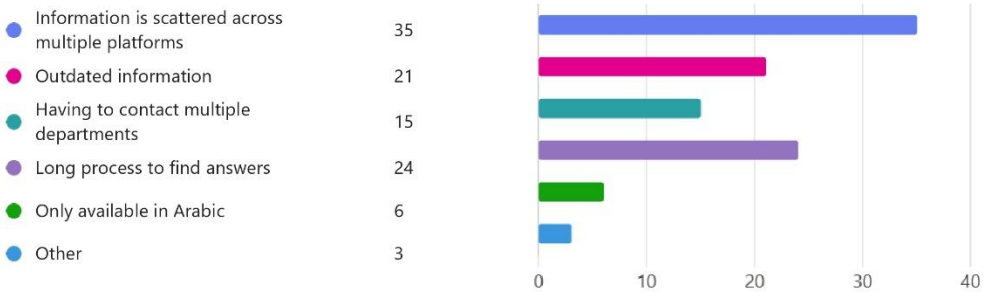


2. Which year are you in?

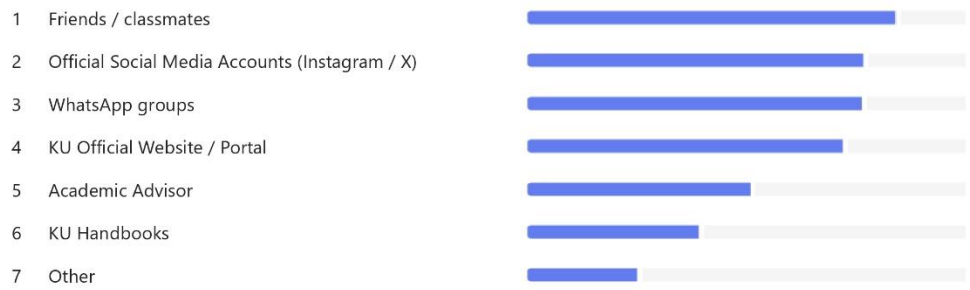


3. How often do you struggle to find accurate or updated university information?

4. What challenges do you face when searching for university information?



5. When you need academic or administrative information, where do you usually look first? *(Rank from most to least used)*



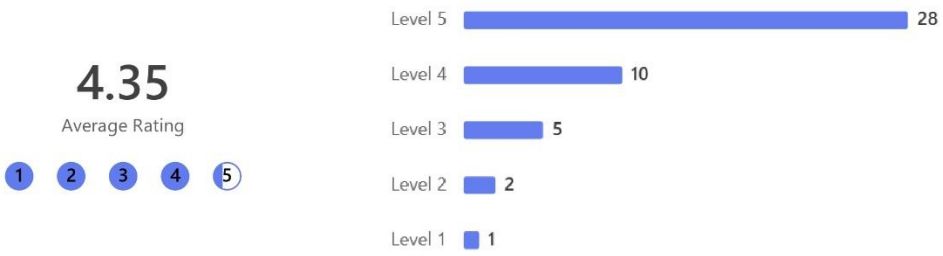
6. If you use other source not listed above, please specify:

12
Responses

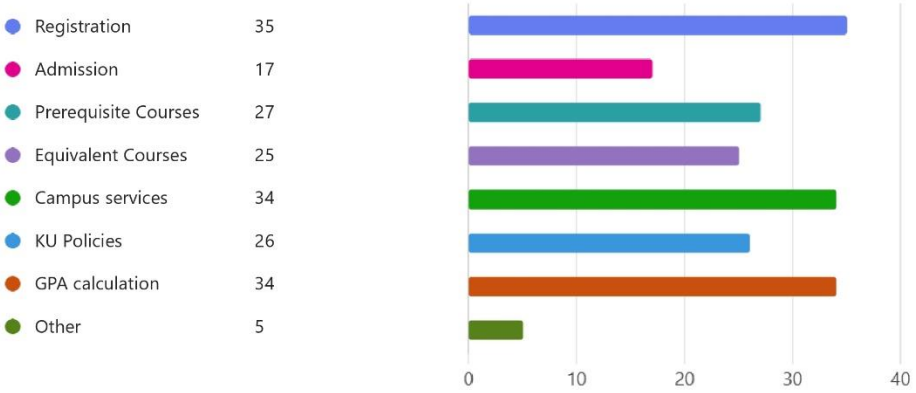
Latest Responses
"ثابت جامعة الكويت"
...

2 respondents (17%) answered None for this question.

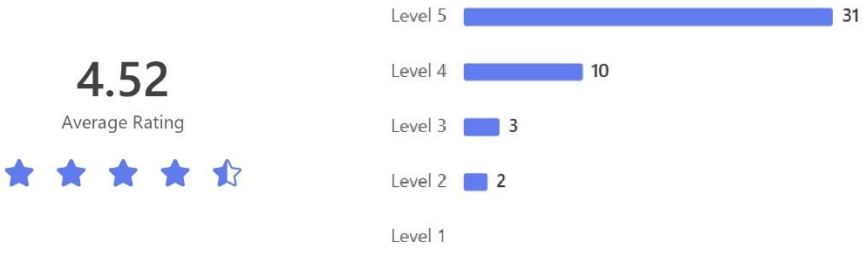
7. If there was a chatbot to answer your university-related questions, how often would you use it?

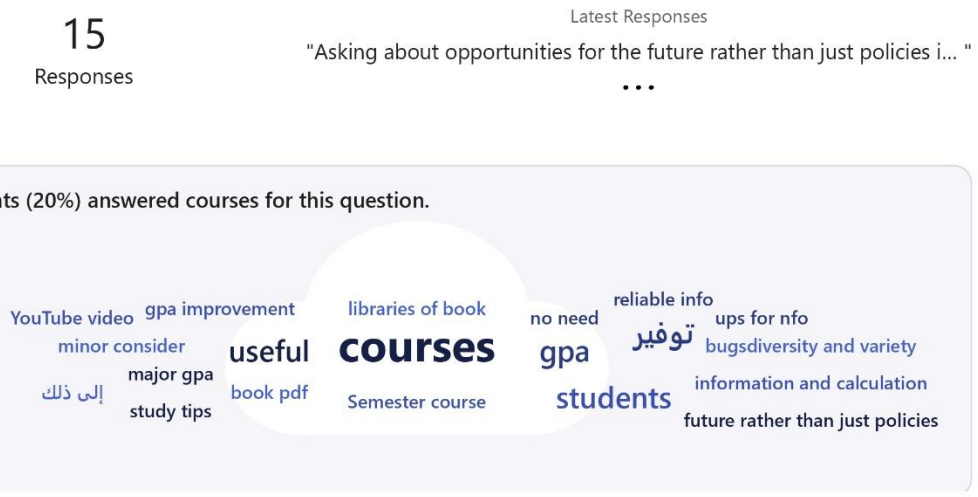


8. Which topics would you most want the chatbot to answer?



9. If the chatbot can estimate your GPA by just entering your grades, would you find this feature useful?



10. How important is it for the chatbot to show the source of its information?

11. Do you have any other suggestions or features you would like to see in the chatbot?